

EE 132 Midterm

Spring 2026

Student Name:

Exam rules:

Open-book, open-note exam.

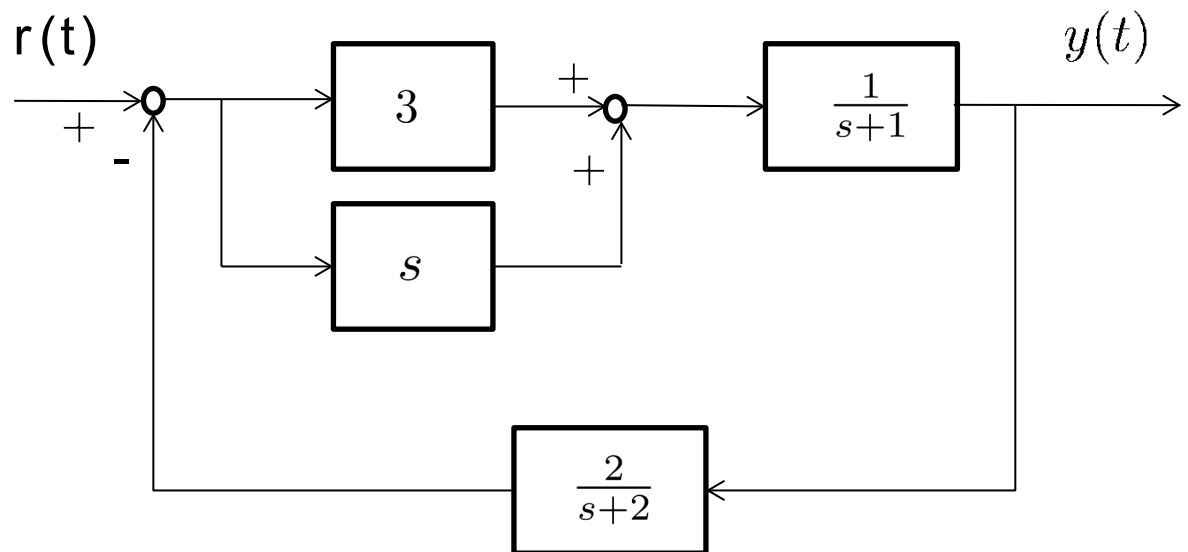
No computer or calculator allowed.

Collaboration is strictly forbidden (You will receive a zero if there is any plagiarism.)

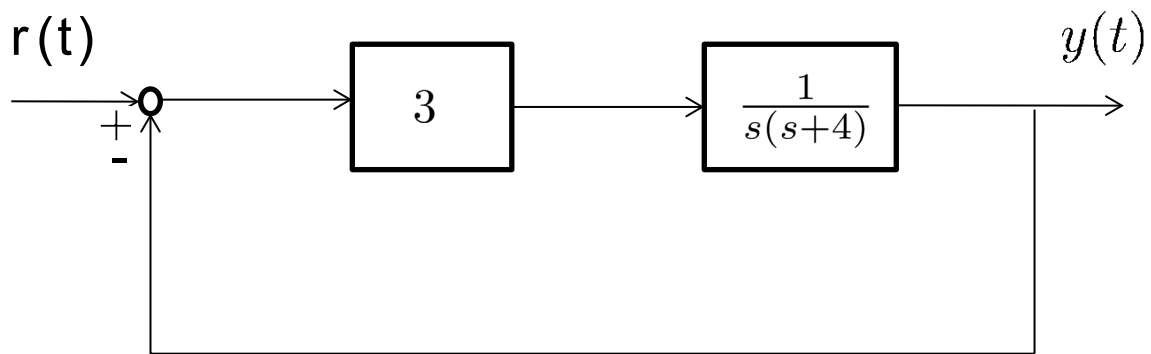
Attempt all problems and write neatly.

1. Given a system modeled by a differential equation $\ddot{y} + 2\dot{y} + 3y = 2\dot{u} + 3u$, find the transfer function $\frac{Y(s)}{U(s)}$.

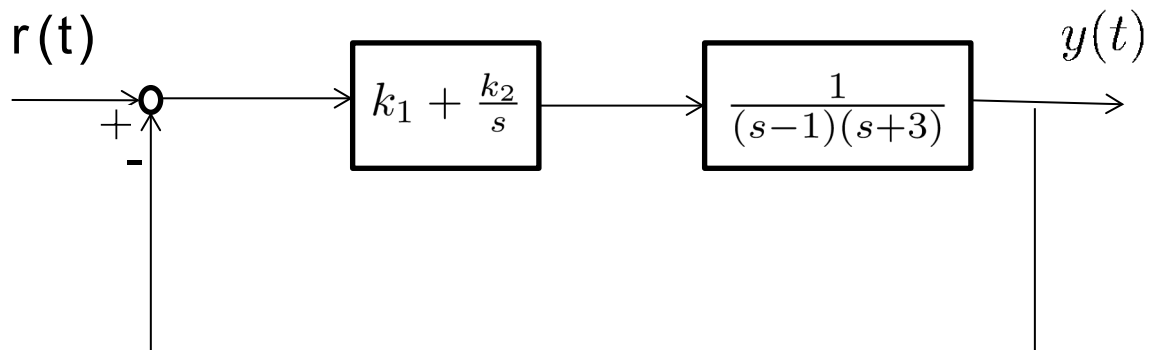
2. What is the closed-loop transfer function of the following system? Let $r(t)$ be a unit-step input. What is the steady-state value of $y(t)$?



3. Suppose that the initial conditions are zero. Let $r(t)$ be a unit step input. What is $y(t)$?



4. Find the range of k_1 and k_2 such that the following closed-loop system is stable. What is the type of the closed-loop system? What is the steady state error when $r(t)$ is a step signal, $k_1=5$ and $k_2 = 1$?



5. Specify the parameters k_1 and k_2 such that the overall closed-loop system response to a unit-step input has a rise time of 1.8 seconds and a 1% settling time of 9.2 seconds.

